

RICH LAKE AGDM, AB, Canada

WMO: 713490

Lat: 54.5012N

Lon: 111.6914W

Elev: 2021

StdP: 13.65

Time Zone: -7.00 (NAM)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-28.8	-22.5	-35.1	0.8	-28.5	-28.4	1.2	-22.1	21.3	24.5	19.4	22.7	4.0	280	0.673

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.0	81.9	64.0	78.5	62.2	75.3	60.8	66.5	78.3	64.2	75.1	62.3	72.4	8.1	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.2	90.3	72.6	59.9	83.1	69.1	58.1	77.7	66.7	32.3	78.4	30.4	75.3	29.0	72.4	74.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.0	16.7	14.8	DB	-37.1	86.9	6.3	3.4	-41.6	89.4	-45.3	91.4	-48.9	93.3	-53.5	95.8	(4)
(5)			WB	-37.3	70.7	6.3	1.8	-41.8	72.0	-45.5	73.0	-49.1	74.0	-53.6	75.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	35.1	8.5	10.1	21.4	36.4	49.3	57.1	61.7	58.4	49.5	37.2	20.9	8.7	(6)
	DBStd	22.32	16.18	14.31	13.55	9.80	8.15	5.64	5.11	5.76	7.91	8.58	12.55	14.94	(7)
	HDD50	6493	1285	1117	886	415	113	10	0	6	103	405	872	1281	(8)
	HDD65	10966	1750	1537	1351	857	489	242	128	214	467	863	1322	1746	(9)
	CDD50	1047	0	0	0	8	90	224	364	267	87	7	0	0	(10)
	CDD65	43	0	0	0	0	1	6	26	9	1	0	0	0	(11)
	CDH74	961	0	0	0	1	66	172	421	225	73	3	0	0	(12)
	CDH80	184	0	0	0	0	9	33	97	31	14	0	0	0	(13)
	WSAvg	6.8	6.5	6.5	7.4	7.9	7.6	6.5	6.1	6.1	6.5	7.1	7.0	6.2	(14)
	PrecAvg	17.2	0.7	0.6	0.7	1.2	1.7	3.0	3.3	2.4	1.4	0.9	0.8	0.7	(15)
(16) Precipitation	PrecMax	20.7	2.0	1.3	1.4	3.4	3.2	6.4	7.2	5.4	3.6	2.2	2.2	1.4	(16)
	PrecMin	10.9	0.0	0.0	0.2	0.1	0.2	0.9	1.3	0.9	0.1	0.2	0.1	0.0	(17)
	PrecStd	2.6	0.4	0.3	0.4	0.7	0.8	1.3	1.1	1.0	0.9	0.4	0.6	0.4	(18)
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.7	44.7	52.3	71.1	80.5	83.7	87.4	83.9	81.7	71.3	51.8	42.5	(19)
		MCWB	37.1	36.5	41.9	51.0	58.0	63.1	67.6	65.2	61.4	54.5	41.8	36.4	(20)
	2%	DB	39.8	39.2	46.9	64.3	74.8	78.9	82.5	79.5	74.8	60.6	44.7	38.2	(21)
		MCWB	34.5	33.6	38.3	47.3	55.2	60.4	66.5	63.6	58.4	48.8	37.7	33.5	(22)
	5%	DB	36.0	34.8	42.8	58.1	70.6	74.5	78.5	76.3	69.4	54.9	40.5	34.0	(23)
		MCWB	32.1	30.6	35.8	44.0	52.7	58.4	63.8	62.6	56.2	44.9	35.5	30.2	(24)
	10%	DB	31.3	30.7	38.7	52.9	66.6	71.0	75.1	72.7	64.1	50.5	36.3	29.3	(25)
		MCWB	28.6	27.7	33.2	41.4	50.5	56.9	62.4	60.6	53.5	42.6	32.7	26.8	(26)
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.4	37.2	42.4	52.4	60.0	65.5	70.4	68.7	62.7	54.8	43.1	36.9	(27)
		MCDB	43.0	42.9	51.4	69.8	75.4	80.5	82.0	79.2	78.2	68.4	50.9	41.8	(28)
	2%	WB	34.9	33.9	38.7	48.2	56.3	62.7	67.7	65.4	59.5	49.5	38.9	33.8	(29)
		MCDB	39.2	38.7	45.9	62.8	70.9	74.8	79.4	76.3	72.5	58.4	43.9	38.2	(30)
	5%	WB	32.4	30.9	36.1	44.6	54.2	60.6	65.4	63.2	56.9	45.9	35.8	30.4	(31)
		MCDB	36.1	34.5	42.2	57.0	67.8	71.4	76.5	73.7	68.0	53.5	39.9	33.7	(32)
	10%	WB	28.6	27.8	33.7	42.0	51.9	58.4	63.4	61.3	54.4	43.3	32.7	26.9	(33)
		MCDB	31.5	30.9	38.1	51.7	64.1	68.1	73.0	71.3	62.8	49.5	35.9	29.3	(34)
(36) Mean Daily Temperature Range	5% DB	MDBR	17.9	20.1	19.8	20.8	24.2	22.2	22.0	22.4	21.8	18.0	15.2	16.9	(35)
		MCDBR	19.4	23.8	23.2	30.8	31.5	29.6	28.5	29.4	32.2	27.0	19.2	21.2	(36)
		MCWBR	15.0	19.1	15.7	17.0	15.2	14.2	14.2	15.1	16.9	16.2	13.5	17.4	(37)
	5% WB	MCDBR	19.4	23.5	22.2	29.1	28.1	25.7	26.4	27.0	30.1	25.8	18.5	20.8	(38)
		MCWBR	15.3	19.0	15.3	16.7	14.3	13.5	14.0	14.6	16.2	16.0	13.6	17.5	(39)
(40) Clear-Sky Solar Irradiance	taub		0.251	0.260	0.288	0.325	0.342	0.356	0.358	0.363	0.310	0.280	0.252	0.234	(40)
	taud		2.271	2.315	2.356	2.394	2.402	2.373	2.423	2.354	2.517	2.574	2.449	2.281	(41)
	Ebn at Noon		227	268	285	287	286	282	280	270	275	262	232	210	(42)
	Edh at Noon		18	24	29	33	34	36	34	34	25	19	15	14	(43)
(44) All-Sky Solar Radiation	RadAvg		269	597	1081	1476	1797	1849	1873	1527	1068	593	275	188	(44)
	RadStd		29	48	76	121	153	127	101	120	113	95	35	18	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only			-1.62	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (51 neighbors)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page